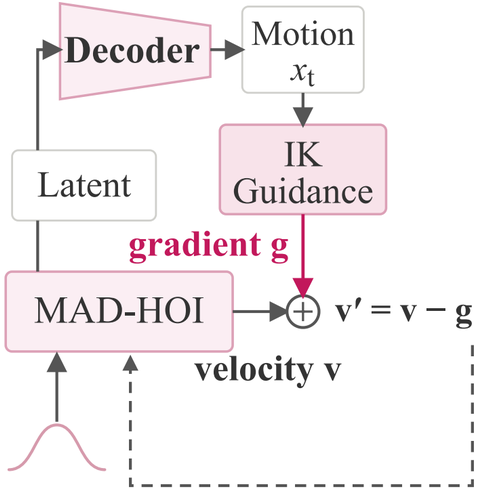
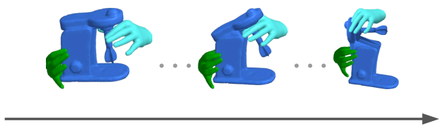
1 IK Guidance at Inference



Iterative energy guidance with differentiable IK modules

2 Guided Motion Generation



Generated Trajectory (MANO)

- 2 × Hand: **3D trans.** + **6D rot.**
- Object: **3D trans.** + **6D rot.** + **1D** articulation

3 Retargeting

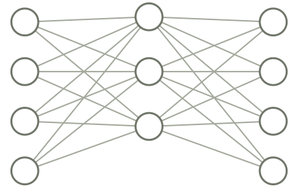


Reference Motion

15 DoF upper body + 24 DoF (hands)

- **Optimized robot base**
- **Lower kinematic energy**

4 RL Agent



MLP Policy

Decoupled arm + hand rewards

Decoupling increases SR by 3.1%

5 Sim Rollout



Task SR increases 65% → 72% over unguided samples